

EDUCATIONPLUS

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Mamidala Jagadesh Kumar

India's higher education system faces two challenges at once. A qualification earned in one part of the country should be recognised and respected elsewhere but the education offered must also be relevant to its students, languages, social setting, and surrounding economy. These aims may appear to pull in different directions. In practice, they can reinforce each other.

This balance is the essence of the National Education Policy (NEP) 2020, which makes responsiveness to India's diversity and local conditions a guiding principle for curriculum, teaching, and policy. It calls for careful planning, joint monitoring, and collaboration between the Centre and the States, supports institutional autonomy and the expansion of programmes in local languages, and provides meaningful room for State priorities and local aspirations.

Guidelines

National frameworks provide broad guidelines to the institutions without imposing rigid prescriptions. For instance, the UGC framework leaves important academic decisions to institutional statutory bodies. Boards of Studies and Academic Councils can determine course lists and allocate credits across different programme structures.

Build the right connections

Strengthening institutional capacity, measuring outcomes reliably and using better regional data can create more opportunities for all learners



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rect" course based on short-term market trends. The All India Survey on Higher Education (AISHE) 2023-24 records 4.50 crore students, compared with 3.42 crore in 2014-15. Over the same period, the gross enrolment ratio for the 18-23 age group rose from 23.7 to 30. These gains indicate an expanding social base. They are useful for building measures for flexibility, inclusion, and regional capacity.

Beyond course preferences, institutional reputation will continue to influence student decisions. Reputation may reflect teaching and research, but also history, location, alumni, programme range, and public visibility. A constructive approach is to widen the number of institutions that students and families can choose with confidence.

Student mobility

At the same time, local relevance should not confine a learner to the place of study. The National Credit Framework and the Academic Bank of Credits seek to make academic, vocational and experiential learning recognisable and portable across institutions and States. Institutions must ensure that credits are accurately recorded, transferred, and redeemed to enable students to complete their programmes promptly. When these arrangements work well, portability and place-based learning support student mobility.

A related question con-

cerns the relationship between curriculum and employment. Degrees cannot be designed only around immediate job openings because economies and occupations change. Higher education must build knowledge, judgement, communication, creativity, and the capacity to learn. It should also give students opportunities to apply those capabilities in professional, social, and community settings. NEP 2020's emphasis on multidisciplinary learning, internships, vocational exposure and local opportunity mapping brings these purposes together.

The UGC's guidance enables institutions to develop internship projects suited to their academic programmes and the surrounding economic environment. Its university-industry framework encourages regional clusters, Industry Relations Cells and engagement with micro, small, and medium enterprises. Apprenticeship-embedded degree programmes and Professors of Practice provide further channels for professional participation. AICTE's model curricula and applied-learning initiatives similarly connect technical education with projects, internships and emerging fields. These mechanisms respect the academic role of institutions while bringing employers and practitioners into appropriate parts of the learning process.

Accordingly, we should continue to build a stron-

ger evidence base linking education to opportunity. From graduating students, institutes could collect information on employment, further study, entrepreneurship, location, wages, and the use of skills at suitable intervals after graduation, while respecting privacy issues. They should also present student-facing programme information, such as curriculum, costs, eligibility, accreditation, internships, completion, and possible pathways, in clear language, without making predictive promises. States and regional skills centres could bring together official labour data, apprenticeship information, industry evidence and institutional records. This can help students and families, institutions, employers and States make better decisions.

Ultimately, India needs a connected education system in which qualifications are recognised across regions. Institutions should adapt to local needs, while students should be free to choose their learning pathways. We should continue to strengthen institutional capacity, measure outcomes reliably and use better regional data. NEP 2020 provides us with a path to develop national frameworks that focus on local relevance to create better opportunities for all learners.

Views are personal

The writer is the Chairman of the Review Committee for NEP 2020 and Chairman, BoG, IIM-Calcutta.

SCHOLARSHIPS

Nikon Scholarship

An initiative by Nikon **Eligibility:** Indian citizen who has passed class 12 (any stream) and is currently pursuing a professional photography-related course of minimum three-month duration and an annual parental income not exceeding ₹600,000 **Rewards:** Based on actual course fee. **Application:** Online **Deadline:** August 24 www.b4s.in/edge/NIKONI4

CSIR-NIScPR Internship

Offered by the CSIR-NIScPR **Eligibility:** Indian citizens pursuing a UG, PG, or doctoral programme in a recognised institution and demonstrate interest in science communication, policy research, or a related interdisciplinary field. **Rewards:** Certificate **Application:** Online **Deadline:** Round the year www.b4s.in/edge/NISCI1

IIRS External Student Internship Programme

Offered by the IIRS **Eligibility:** Indian nationals who have a UG, PG, or Ph.D. degree in Science or Technology from recognised institutions with minimum 60% aggregate. **Rewards:** Certificate **Application:** Online **Deadline:** October 30 www.b4s.in/edge/IIRS2

Courtesy: Buddy4study.com

Shift the focus

Uncertain about your career options? Low on self-confidence? This column may help

do 20 high-quality MCQs than rush through 100 passively. Finally, a skipped day isn't a failure. Reset and hit your daily minimum target the next morning.

I am in Class 10 and want to pursue Maths for higher studies. But I do not want to continue with Science subjects in Class 11 and 12. What else can I do so that I can study Maths later? Palash

Dear Palash, There is a misconception that if you want to study higher-level Maths, you must choose Science. Maths is also deeply intertwined with Economics, Commerce, Data Science, and the Social Sciences.

After Class 10, you can choose Commerce or the Humanities and select Maths as an elective. Commerce and Maths open doors to finance, actuarial science and quantitative fields, while Humanities and Maths are useful for public policy, demographic research and economics.

CBSE offers Core Maths covering calculus, trigonometry and algebra, and Applied Maths focussed on business, economics, finance and data analysis. Check the eligibility criteria for your preferred degree before choosing. After Class 12, you can pursue degrees through CUET, including B.Sc./B.A. Mathematics, Actuarial Science, Economics, Data Science and Statistics.

Modern Economics is highly mathematical, while Data Science combines Maths, programming and analytics. Choose the stream that matches your interests.

Disclaimer: This column is merely a guiding voice and provides advice and suggestions on education and careers.

The writer is a practising counsellor and a trainer. Send your questions to eduplus.thehindu@gmail.com with the subject line Off the Edge

India and abroad with a focus on research and teaching careers? What are the other career options? Lena

Dear Lena, Environmental Science relies on mathematical modelling of climate systems, physical principles of fluid dynamics, chemical analysis of pollutants, and biological insights into ecosystems. Your combination of Biology and Maths is ideal. Since your goal is research and academia, target premier institutes that emphasise experimental design, field studies, and data analysis. In India, the gold standard is an integrated BS-MS or Bachelor of Science (Research) programme.

The Indian Institute of Science, Bengaluru, offers a four-year B.Sc. (Research) programme with a major in Earth and Environmental Science. IISERs across Pune, Kolkata, Mohali, Bhopal, and Thiruvananthapuram have outstanding Earth and Climate Sciences departments. IIT Bombay offers an Interdisciplinary Programme in Climate Studies, while IIT Kharagpur has a five-year integrated M.Sc. in Applied Geology and Earth Sciences. JNU and BHU are strong options for postgraduate research.

If you aim to study abroad, consider Stanford, Columbia, and UC Berkeley in the U.S., Oxford and Cambridge in the U.K., ETH Zurich in Switzerland, and Wageningen University in the Netherlands.

High-demand niches in academics include Climate Modelling and Simulation, Biogeochemistry, Palaeoclimatology, and Satellite Remote Sensing

and GIS. Alternative careers include corporate sustainability, think tanks, and international bodies such as UNEP, the World Bank, IPCC and WWF, and the green technology industry.

I am in the third year of medical college and under pressure to crack NEET-PG in the first attempt. I face problems with consistency and time-management. How can I overcome this? Shreeya

Dear Shreeya, You are starting your focused preparation at the perfect time. You can now build your core clinical concepts without the hectic schedule of a final-year internship. Shift from a 'time-based' to a 'task-based' approach. Hospital schedules and exhaustion can break your schedule. So, use micro-targets. Commit to 20-30 high-yield MCQs and three video modules/textbook topics per day. Utilise 'dead time' and keep your Q-Bank app active.

Focus on mastering your third-year subjects while maintaining a baseline touch with previous years. Ophthalmology, ENT and Community Medicine (PSM) account for a significant chunk of the exam. Spend weekends revising Pathology, Pharmacology and Microbiology from second year.

Break the cycle of procrastination with mini-tests and mock exams. Solve 10-15 MCQs, followed by 45-50 minutes of Targeted Review. Bookmark, tag, and flag volatile facts for weekly review. Ignore the 'Herd Noise'. Follow your path. Perfectionism is the enemy of consistency. It is better to



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Chance to explore

Why summer schools are the missing link between school and university

Vinnie Mathur

Every year, thousands of students find themselves facing one of the first major decisions of their lives: what to study after school. They compare universities, browse course brochures and speak to family, teachers, and friends. Yet, many make these choices without ever experiencing what those subjects actually look like beyond the classroom.

More options

This challenge has become even more relevant now, as students have more academic options than ever before. Interdisciplinary learning is also becoming increasingly common, allowing students to combine different interests and build unique academic journeys. The National Education Policy (NEP)

2020 has further accelerated this shift by encouraging multidisciplinary learning and greater flexibility in academic pathways. But all this also raises an important question. How can students make informed decisions about fields they have never had the opportunity to experience?

This is where summer schools are beginning to play a far more meaningful role than they did in the past. Traditional summer schools were viewed as a way for students to learn a new skill, pursue a hobby or stay productively engaged during their summer break.

Now, however, they have the potential to become spaces where students can explore varied disciplines, interact with experts and gain a first-hand understanding of university-style learning before making important aca-

demic decisions. Interestingly, most students who join these programmes are not looking for certainty but for clarity. They want to test an interest, discover something new or simply understand whether a particular subject excites them enough to pursue it further.

Option to explore

That exploration often begins with a simple change in the learning environment. A design challenge, a laboratory experiment or a discussion on a contemporary issue can reveal dimensions of a subject that are difficult to capture within the structure of a school curriculum. Students quickly realise that learning at the university level is often less about memorising answers and more about questioning assumptions, testing ideas and connecting knowledge

across disciplines.

Equally valuable is the opportunity to experience university life before becoming part of it. Questions that begin with admissions requirements or career prospects often evolve into discussions about research, innovation, emerging technologies and the future of work. Summer schools also allow students to engage with current university students, visit laboratories and experience campus life first-hand.

Another important advantage is the opportunity to learn alongside peers from different schools, cities, and backgrounds. Conversations during group projects, meals and informal interactions often end up presenting these students with a new looking glass, and letting them think beyond their existing purview even after the programme ends.

As higher education becomes increasingly interdisciplinary and dynamic, helping students navigate the transition from school to university has never been more important. Summer school allows exploration, encourages curiosity, builds confidence, and allows students to ask better questions about their interests, abilities and aspirations. In many ways, that may be its greatest contribution. Not helping students find all the answers, but helping them discover the questions that matter most before they take their next step.

The writer is Chair of the Young Thinkers Forum, Shiv Nadar University Delhi-NCR.



Win for India

Team Reviv3D — Vidushee, Shravya, and Shloka, students of Mallya Aditi International School, Bengaluru — took the first place at the global finals of Monash

University's Change It Challenge in Melbourne.

The team — represented in Melbourne by Vidushee and Shravya — developed affordable, high-quality prosthetics by combining bagasse — the dry, pulpy fibrous material remaining after

sugarcane or sorghum stalks are crushed to extract their juice — and crushed basalt from stone with recycled plastics. This composite material is stronger than many fiberglass alternatives, drastically cheaper, and fully recyclable.

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